Cavaliers 2026 Sports Business and Analytics Hackathon

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Our Approach

We began by exploring relationships across the historical event, audience, and concessions datasets. Our goal was to identify understandable event groups that could support forecasts and pricing decisions for upcoming shows. We hypothesised that combining event characteristics, audience composition, and purchasing patterns would provide more useful predictions than examining any one factor alone.

Alcohol spending was an early point of differentiation. We investigated how it varied alongside event type, genre, and audience demographics, while also examining demand for water, fountain drinks, and pretzels.

Methodology

Initial exploration and group development

We joined the show, audience, and concessions datasets to investigate why purchasing patterns differed between events. Our initial observations suggested that Family and Sports events had lower prices for some concessions and a stronger emphasis on food and non-alcoholic drinks, while Concerts and Comedy events generated more alcohol spending. The accompanying “Initial Findings and Ideas” file documents our exploratory process.

We then examined genre and audience demographics. Differences in male ticket-buyer share helped inform a split within Family events. This pattern of behavior was exhibited in multiple Family events and led to four initial groups.

Family Day included Family events with less than 50% male ticket buyers, while Family Night & Sports combined Sports events with Family events whose male ticket-buyer share was at least 50%. Bar comprised Rock and Country concerts, and Social covered the remaining events, including Pop, Hip-Hop / R&B, and Comedy.

Testing and reducing prediction error

We used backtesting to compare predicted concession units with observed historical sales. Early aggregate results prompted us to examine individual items and use several validation methods, rather than relying on a single error figure.

With AI-assisted coding and testing, we expanded the comparison to 260 distinct grouping schemes, ranging from one to five groups. We evaluated each scheme using three validation designs: Performer holdout, exclude every show by one performer, then predict those shows. Show holdout, exclude one show at a time; other shows by that performer may remain in training. Repeated performer-based five-fold validation: Divide performers into five sets, hold out each set in turn, and repeat the process 20 times.

We assessed predictions using Weighted Absolute Percentage Error (WAPE), which measures total absolute error relative to total units sold; sMAPE, which averages symmetric percentage errors across show-item predictions; and normalized RMSE, which gives greater weight to large misses.

Final group selection

Our original structure with Pop separated into a fifth group performed best across all three error measures and validation designs.

We divided the 36 historical shows into five groups: Family Day included the seven remaining Family events after assigning Family events with at least 50% male ticket buyers, alongside Sports events, to Family Night & Sports (five shows). Bar comprised nine Rock and Country concerts, Pop included five Pop concerts, and Social covered ten Hip-Hop / R&B and Comedy events.

Key Findings

The five-group model reduced performer-holdout WAPE from 23.78% to 21.18%, an improvement of 2.60 percentage points, or approximately 11% less error, compared with the reconstructed four-group baseline. These figures use each group's total units divided by its total tickets, as our forecast does; with an average over each group's shows instead, as in our visual, the same test gives 24.0% and 21.8%. Earlier item-level comparisons indicated that the improvement mainly came from water, fountain drinks, and pretzels. The five groups also buy very differently: Bar crowds spend about 7 times more on alcohol per fan than Family Day crowds ($13.30 vs $1.80), Pop fans buy the most water (168 bottles per 1,000 fans), and Social crowds (hip-hop/R&B and comedy) buy more cocktails than Pop crowds (71 vs 48 per 1,000 fans).

Roadblocks

Manual exploration was time-consuming, so we used AI tools to accelerate coding and testing while retaining responsibility for interpreting the results. With only 36 historical shows, some groups had limited supporting observations.

The tests used actual final ticket counts, so the reported errors measure concession demand accuracy given a known crowd size. Forecasting attendance and estimating responses to price changes introduce additional uncertainty.

The Pop split also emerged during earlier exploration of the same data. Testing many schemes on these shows may favour patterns specific to this sample, making future-event validation important. Two checks support the simple rule: in a bootstrap over performers, the Pop split lowered the error in 98% of resamples, and versions tuned on the same data (demographic adjustments, a different method for each item) did worse once the choice was tested on acts it had not seen (nested validation). Demographics describe ticket buyers rather than concession purchasers, and the concessions records cover most, but not all, points of sale.

Recommendations

Here are our final prices that we have come up with. The 20oz Dasani water's new price should be $4.99 for every show. Water is a necessity and should not be overpriced regardless of the event. For Family Day and Family Night & Sports events, new prices are cocktail $22.99, fountain $6.49, pretzel $7.49, Michelob $12.99. Revenue is smaller for Family events so keeping concessions affordable has more of a positive impact than a small gain. For Pop and Social events, cocktails increase to $30.25 and Michelob to $17.49. Bar events see the highest increase, with cocktails at $30.99 and Michelob at $17.99. Fountain drinks ($7.79) and pretzels ($10.59) keep today's prices at these shows. At a higher alcohol consumption rate, and higher spend per ticket, we can assume consumers are less price sensitive to the increase in price. At these prices, concessions are kept cheap when there is an abundance of family buyers and volume and prices only increase when the crowd cares least for the price.

For the 9 upcoming shows at the base crowd, we forecast $221,923 in gross sales at these prices vs $220,868 at the old prices (+$1,055), with 4.0% more items sold. Replayed on last season's 36 shows with the same assumptions, they would have added $8,682 (+0.6%) and 3.3% more items. The new prices stay ahead at every price sensitivity we tested (0x to 2x the published values), while final attendance moves the total much more ($195K to $276K across our low and high crowd cases). Our visual (Show_Planner.pdf, also live at zeynepbastas-cavs-show-planner.streamlit.app) shows the group for every historical and upcoming show and the projected units, recommended price and projected gross for each item at each upcoming show. The same numbers are in event_groups.csv and upcoming_forecast.csv.

Next steps: place each new show in its group when tickets go on sale and set that group's prices at the stands; use the projected units to plan stock and staffing; and, because every past show used the same menu prices, test the new prices at a few shows in each group and update the price sensitivities with what fans actually buy.